

ISLAM KASSEM

AI / ML Engineer | GenAI & LLM Specialist

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PROFESSIONAL SUMMARY

AI/ML Engineer with 5+ years designing and deploying production AI systems across LLMs, RAG, NLP, Computer Vision, and ASR. Led end-to-end GenAI and MLOps delivery for Saudi government ministries and enterprise clients — including on-premise LLM deployments (Kubernetes), CI/CD-backed model pipelines (GitHub Actions, MLflow/W&B), and cloud-hybrid infrastructure (AWS/Azure). Kaggle Master (top 1%), Top-Rated on Upwork, active open-source contributor (github.com/elcaiseri), MSc Data Science candidate.

CORE TECHNICAL SKILLS

GenAI / LLMs	RAG Pipelines · LLM Fine-tuning (LoRA/QLoRA) · On-premise LLM · LangChain · LlamaIndex · OpenAI API · AI Agents
NLP	Transformers · BERT · GPT · Text Classification · NER · Semantic Search · Sentence-Transformers · spaCy
Computer Vision	Object Detection (YOLO) · Image Search · AutoTagging · CLIP · OpenCV · FAISS visual embeddings
Speech / ASR	Whisper · wav2vec2 · Real-time Speech-to-Text Pipeline Design
ML / DL	PyTorch · TensorFlow · Hugging Face · Scikit-learn · LoRA/QLoRA · Hyperparameter Optimization
MLOps	MLflow · Weights & Biases · Docker · Kubernetes · FastAPI · GitHub Actions CI/CD · Model Monitoring
Cloud / Infra	AWS (EC2, S3, SageMaker) · Azure AI · GCP Vertex AI · On-premise GPU · Vector DBs (FAISS, Chroma, Pinecone)
Data / Lang	Python (expert) · SQL · Pandas · NumPy · Feature Engineering · ETL Pipelines

PROFESSIONAL EXPERIENCE

AI Engineer | Master Works — Nabeh AI

Nov 2025 – Present • Riyadh, Saudi Arabia

- ▶ Architected enterprise RAG pipelines (FAISS/Chroma + on-premise LLM) enabling citation-grounded document Q&A over thousands of records for Saudi enterprise clients.
- ▶ Led on-premise LLM deployment (LLaMA, Mistral) via Kubernetes-orchestrated GPU inference for air-gapped, data-sovereign GenAI environments.
- ▶ Shipped production ASR pipelines (Whisper) processing thousands of audio records monthly, feeding structured NLP extraction workflows.
- ▶ Established MLflow + W&B experiment tracking and GitHub Actions CI/CD, reducing model delivery cycles and deployment errors significantly.
- ▶ Collaborated cross-functionally with product, data, and infrastructure teams to scope and deliver GenAI solutions aligned with Saudi Vision 2030.

AI / ML Engineer | Velents

Jul 2024 – Aug 2025 • Riyadh, Saudi Arabia

- ▶ Delivered AI hiring tools (ASR, video analysis, multi-modal NLP) deployed across 5+ Saudi government ministries, processing thousands of candidate interactions.
- ▶ Engineered transformer-based MoM automation (BERT/T5 summarization + action-item NLP extraction) eliminating manual documentation overhead across enterprise teams.
- ▶ Built multi-modal assessment pipeline (OpenCV + Whisper + NLP) and anti-fraud detection models for government-scale digital recruitment.
- ▶ Led CI/CD integration (GitHub Actions) for ML model versioning and automated retraining, improving deployment reliability and iteration speed.

Machine Learning Engineer | TwentyToo

Aug 2023 – Jul 2024 • Cairo, Egypt

- ▶ Built semantic AI search engine combining CLIP visual embeddings and NLP text features for intent-aware fashion product discovery across millions of catalog items.
- ▶ Trained YOLO-based object detection models for automated fashion item localization, accelerating catalog processing pipelines.
- ▶ Engineered AutoTagging classification system across multi-attribute taxonomy; tracked all experiments with MLflow and W&B.
- ▶ Implemented FAISS-indexed Image Search enabling sub-second visual similarity matching across the product catalog.

AI & ML Engineer (Freelance — Top Rated) | Upwork

Apr 2021 – Aug 2023 • Remote

- ▶ Delivered 50+ custom AI/ML projects for global clients (NLP, CV, tabular, time-series, RAG prototypes) — full lifecycle from scoping to Docker-based deployment.
- ▶ Built automated data pipelines and web scraping infrastructure supporting downstream ML workflows for e-commerce and research clients.

Data Scientist & Community Contributor | Kaggle

Oct 2018 – 2021 • Remote

- ▶ Achieved Kaggle Master (Codes, Top 1%), Competitions Expert, and Discussion Expert — Gold and Silver medals across NLP, CV, and tabular domains.
- ▶ Published 5+ educational ML notebooks cited and forked by thousands of practitioners globally.

KEY PROJECTS

Enterprise RAG Platform — Master Works

Stack: LangChain, LlamaIndex, LLaMA/Mistral (on-premise), FAISS/Chroma, Kubernetes, Docker, GitHub Actions

Retrieval-augmented generation platform for private enterprise document Q&A, deployed in air-gapped Saudi environments. Kubernetes-orchestrated GPU inference ensured production reliability with full data sovereignty.

AI Hiring Intelligence Platform — Velents

Stack: Whisper ASR, Transformers, OpenCV, FastAPI, Docker, GitHub Actions CI/CD

Multi-modal AI system (video, ASR, NLP) for automated candidate assessment deployed across 5+ Saudi government ministries, processing thousands of recruitment interactions with CI/CD-backed model updates.

Fashion AI Search Engine — TwentyToo

Stack: CLIP, YOLO, FAISS, PyTorch, OpenCV, MLflow, W&B

End-to-end visual and semantic product search combining multi-modal embeddings and object detection for fashion e-commerce discovery at scale.

LangChain MCP Adapters #380 — Open Source Contributor

Stack: LangChain, MCP (Model Context Protocol), Python, Async IO, GitHub, CI/CD

Contributed to LangChain MCP Adapters by improving multi-server MCP resource handling and standardizing resource aggregation into LangChain-compatible abstractions for more reliable agent tool and context integration.

EDUCATION

MSc, Data Science | Cairo University, Faculty of Graduate Studies for Statistical Research (Feb 2024 – Present)

BSc, Electrical, Electronics & Communications Engineering | Port-Said University, Faculty of Engineering (2016 – 2021)

CERTIFICATIONS & ACHIEVEMENTS

CNN & Sequence Models — DeepLearning.AI • **Kaggle Master (Codes, Top 1%)** • Gold & Silver Medals — Kaggle • **Top-Rated Freelancer — Upwork** • **5+ Published ML Notebooks**

LANGUAGES

Arabic (Native) • **English** (Professional Working) • **Spanish** (Elementary)